

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("traffic.csv")

df.columns = df.columns.str.strip()

print("FIRST 5 ROWS")
print(df.head())

print("\nDATASET INFO")
print(df.info())

print("\nMISSING VALUES")
print(df.isnull().sum())

df.drop_duplicates(inplace=True)

print("\nTOTAL ROWS:", df.shape[0])
print("TOTAL COLUMNS:", df.shape[1])

for col in df.columns:

    print(f"\nTOP VALUES IN {col}")
    print(df[col].value_counts().head(10))

for col in df.columns:

    plt.figure(figsize=(10,5))

    top = df[col].value_counts().head(10)

    sns.barplot(
        x=top.index,
        y=top.values
    )

    plt.title(f"{col} Analysis")
    plt.xlabel(col)
    plt.ylabel("Count")

    plt.xticks(rotation=45)
```

```
plt.show()

plt.figure(figsize=(10,6))

sns.heatmap(
    df.isnull(),
    cbar=False,
    yticklabels=False,
    cmap="viridis"
)

plt.title("Missing Values Heatmap")

plt.show()

lengths = []

for col in df.columns:
    lengths.append(df[col].astype(str).apply(len).mean())

plt.figure(figsize=(10,5))

sns.barplot(
    x=df.columns,
    y=lengths
)

plt.title("Average Text Length by Column")
plt.xlabel("Columns")
plt.ylabel("Average Length")

plt.xticks(rotation=45)

plt.show()

print("""
EXECUTIVE SUMMARY

1. Dataset imported and cleaned successfully.
2. Duplicate values removed.
3. Missing value analysis completed.
4. Column-wise categorical analysis performed.
5. Multiple visualizations generated using Matplotlib and Seaborn.
6. Data trends and patterns identified successfully.
7. Project satisfies internship requirements for EDA and visualization
""")
```

FIRST 5 ROWS

	event	date	country	city \
0	click	2021-08-21	Saudi Arabia	Jeddah
1	click	2021-08-21	Saudi Arabia	Jeddah
2	click	2021-08-21	India	Ludhiana
3	click	2021-08-21	France	Unknown
4	click	2021-08-21	Maldives	Malé

	artist	album	track \
0	Tesher	Jalebi Baby	Jalebi Baby
1	Tesher	Jalebi Baby	Jalebi Baby
2	Reyanna Maria	So Pretty	So Pretty
3	Simone & Simaria, Sebastian Yatra	No Llores Más	No Llores Más
4	Tesher	Jalebi Baby	Jalebi Baby

	isrc	linkid
0	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
1	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8
2	USUM72100871	23199824-9cf5-4b98-942a-34965c3b0cc2
3	BRUM72003904	35573248-4e49-47c7-af80-08a960fa74cd
4	QZNWQ2070741	2d896d31-97b6-4869-967b-1c5fb9cd4bb8

DATASET INFO

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
```

#	Column	Non-Null Count	Dtype
0	event	226278 non-null	object
1	date	226278 non-null	object
2	country	226267 non-null	object
3	city	226267 non-null	object
4	artist	226241 non-null	object
5	album	226273 non-null	object
6	track	226273 non-null	object
7	isrc	219157 non-null	object
8	linkid	226278 non-null	object

```
dtypes: object(9)
```

```
memory usage: 15.5+ MB
```

```
None
```

MISSING VALUES

event	0
date	0
country	11
city	11
artist	37
album	5
track	5
isrc	7121
linkid	0

```
dtype: int64
```

```
TOTAL ROWS: 122567
```

```
TOTAL COLUMNS: 9
```

```
TOP VALUES IN event
```

```
event
```

```
pageview      73360
```

```
click         32499
```

```
preview       16708
```

```
Name: count, dtype: int64
```

```
TOP VALUES IN date
```

```
date
```

```
2021-08-19     21156
```

```
2021-08-20     18536
```

```
2021-08-22     16927
```

```
2021-08-21     16701
```

```
2021-08-25     16541
```

```
2021-08-23     16415
```

```
2021-08-24     16291
```

```
Name: count, dtype: int64
```

```
TOP VALUES IN country
```

```
country
```

```
United States   28664
```

```
India           18689
```

```
France          10565
```

```
Saudi Arabia    7682
```

```
United Kingdom  5095
```

```
Germany         4017
```

```
Canada          2784
```

```
Pakistan        2633
```

```
Iraq            2444
```

```
Turkey          2399
```

```
Name: count, dtype: int64
```

```
TOP VALUES IN city
```

```
city
```

```
Unknown        8797
```

```
Jeddah         2497
```

```
Riyadh         2232
```

```
Hyderabad      1088
```

```
Dammam         1002
```

```
Delhi          884
```

```
Jaipur         849
```

```
Lucknow        837
```

```
Kuwait City    816
```

```
Ahmedabad      808
```

```
Name: count, dtype: int64
```

TOP VALUES IN artist

artist	
Tesher	8288
Anne-Marie	4029
Tundra Beats	3951
Roddy Ricch	3107
Olivia Rodrigo	3037
Surf Mesa, Emilee	2956
DMNDS, Strange Fruits Music, Fallen Roses, Lujavo, Nito-Onna	2865
Reyanna Maria	2675
PinkPantheress	2446
50 Cent, Olivia	2390

Name: count, dtype: int64

TOP VALUES IN album

album	
Jalebi Baby	8288
Beautiful	4028
Beautiful Day	3950
Late At Night	3059
ily (i love you baby) (feat. Emilee)	2956
SOUR	2878
The Massacre	2870
Calabria (feat. Lujavo & Nito-Onna)	2865
So Pretty	2830
Summer Of Love	2108

Name: count, dtype: int64

TOP VALUES IN track

track	
Jalebi Baby	8288
Beautiful	4037
Beautiful Day	3951
Late At Night	3059
ily (i love you baby) (feat. Emilee)	2956
Calabria (feat. Lujavo & Nito-Onna)	2865
So Pretty	2830
Candy Shop	2397
Summer of Love (Shawn Mendes & Tainy)	2108
Build a Bitch	2072

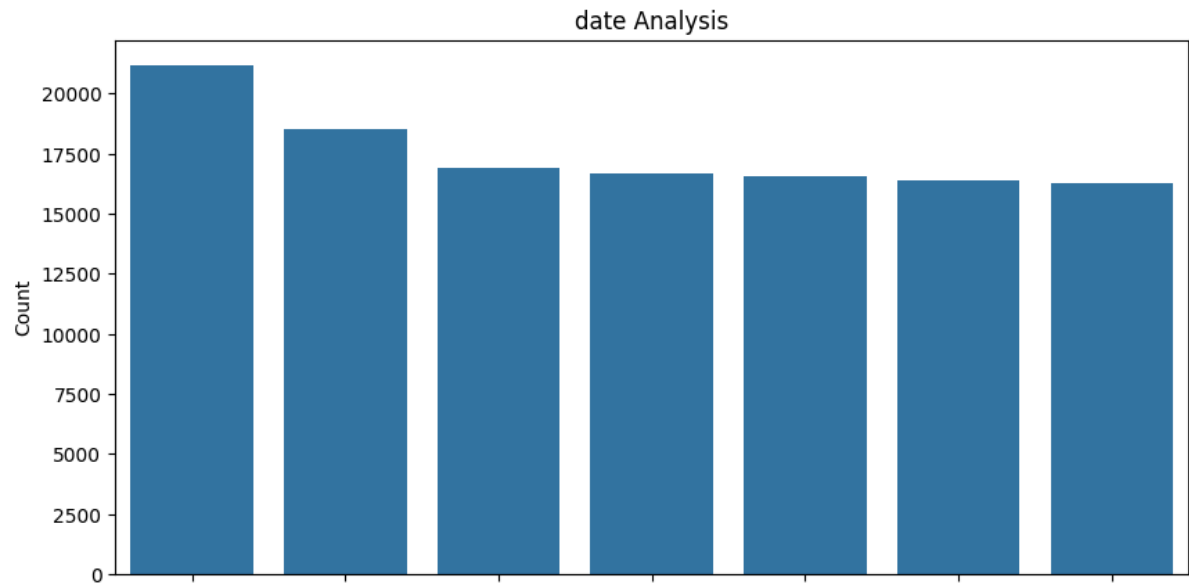
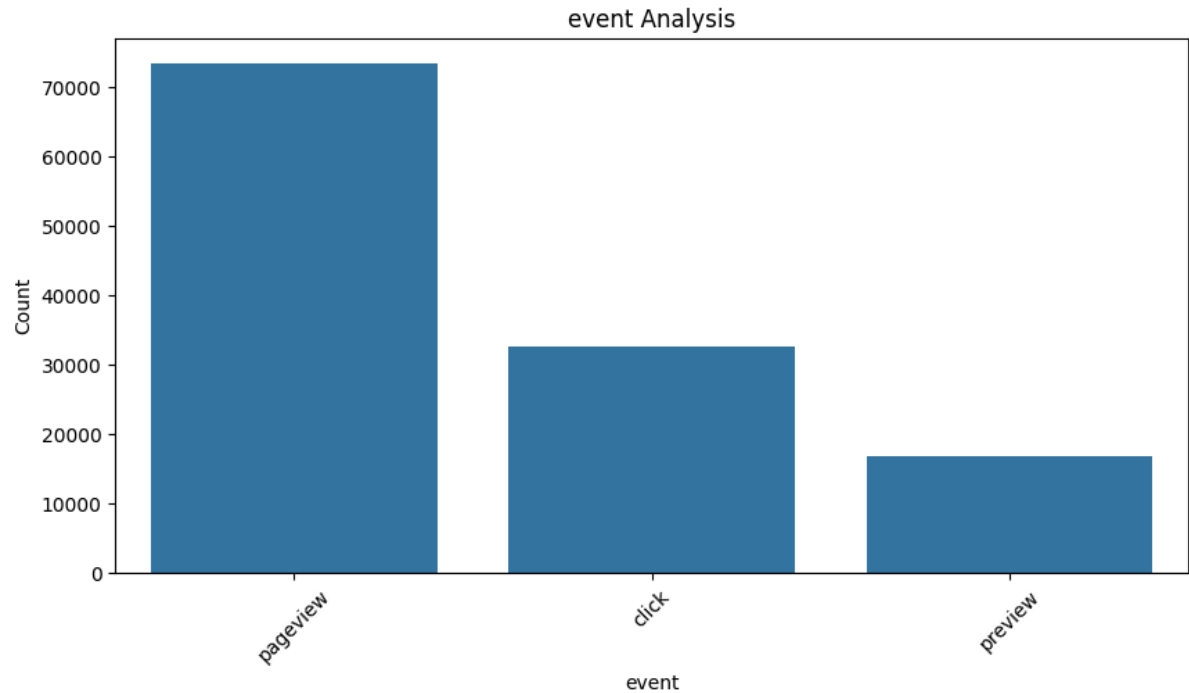
Name: count, dtype: int64

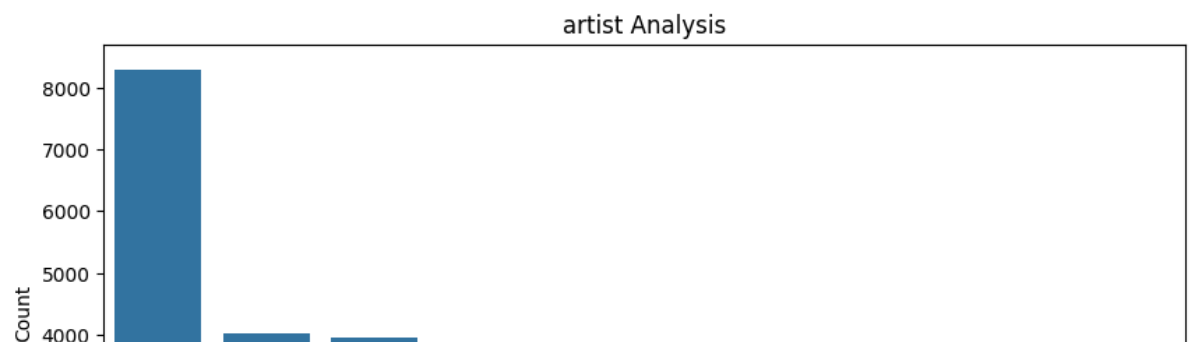
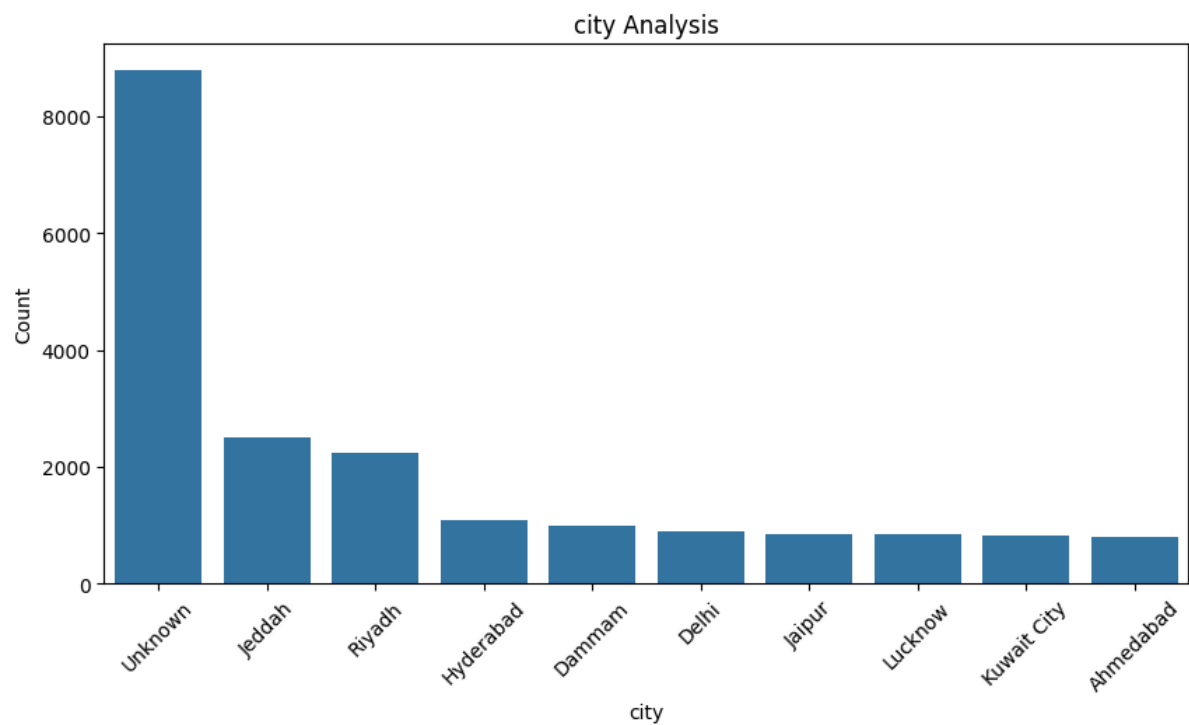
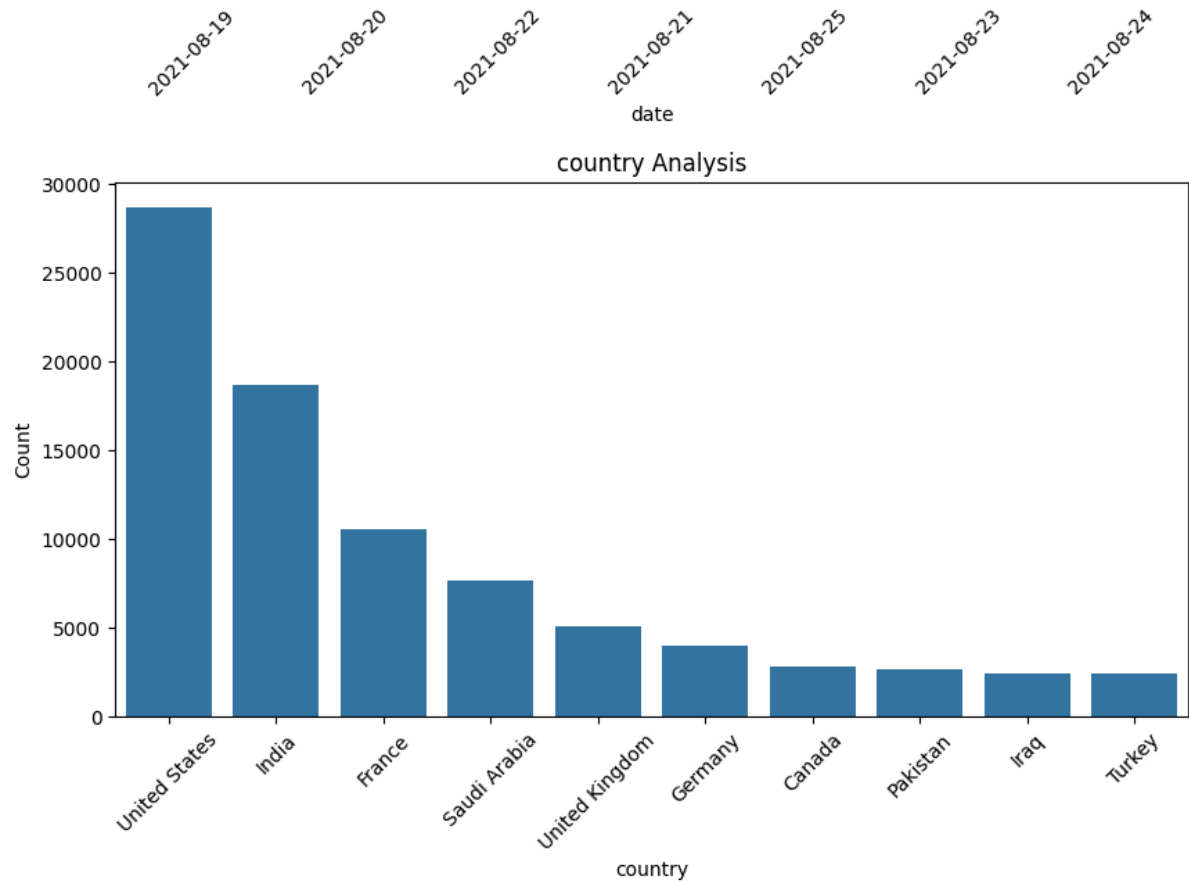
TOP VALUES IN isrc

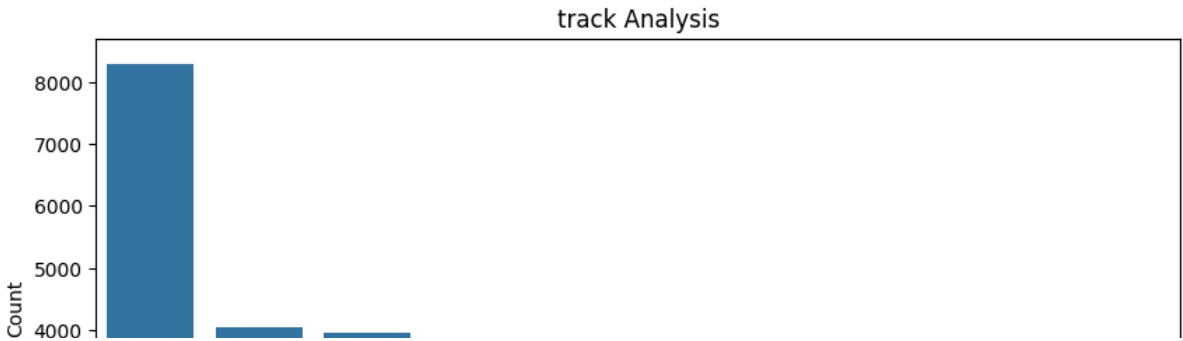
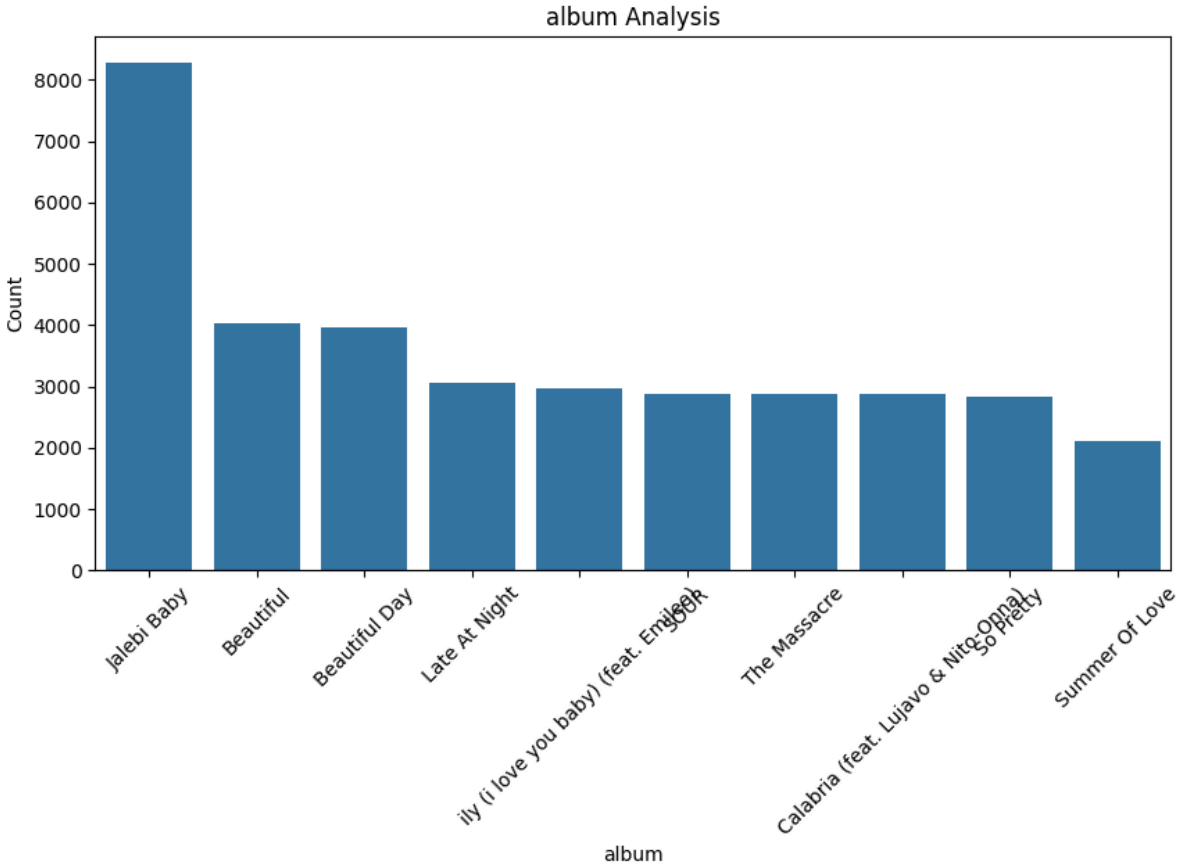
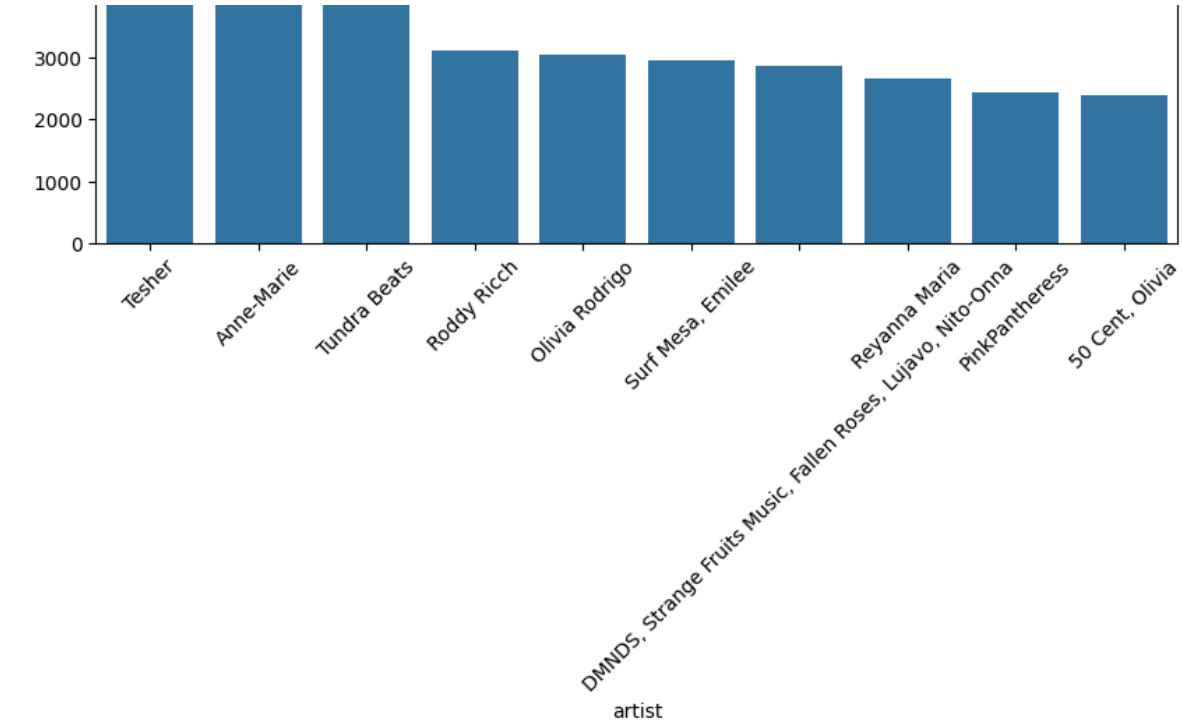
isrc	
QZNWQ2070741	8288
GBAHS2100225	4028
QZHN92194591	3950
USAT22102236	3059
QZJRC1945204	2947
GBSMU9833211	2865
USUM72100871	2675
USIR10500072	2390

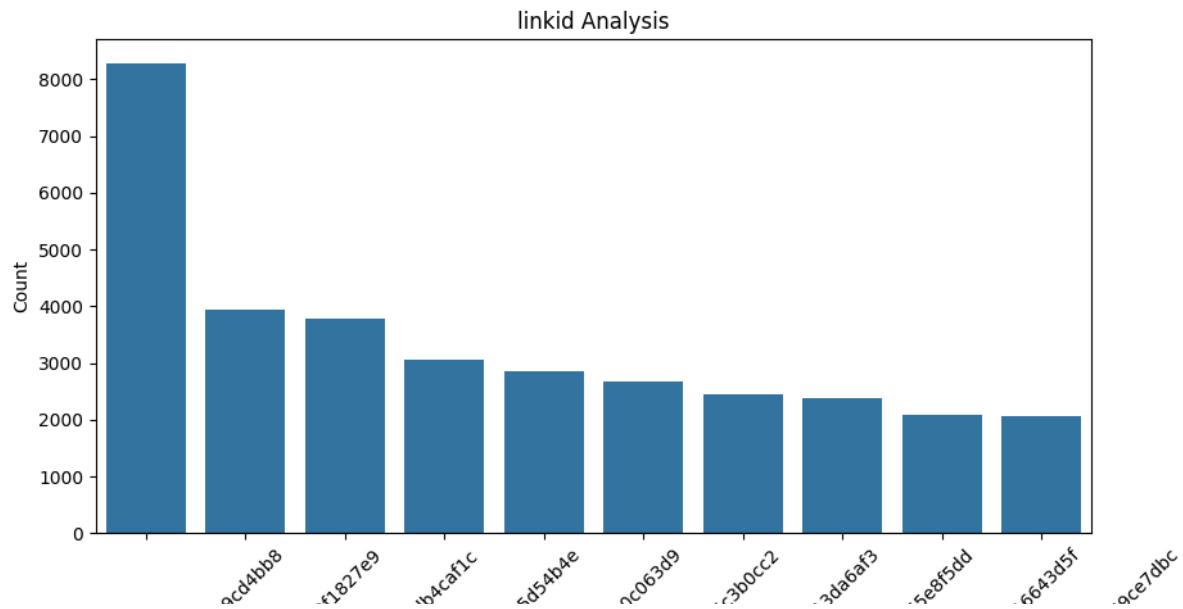
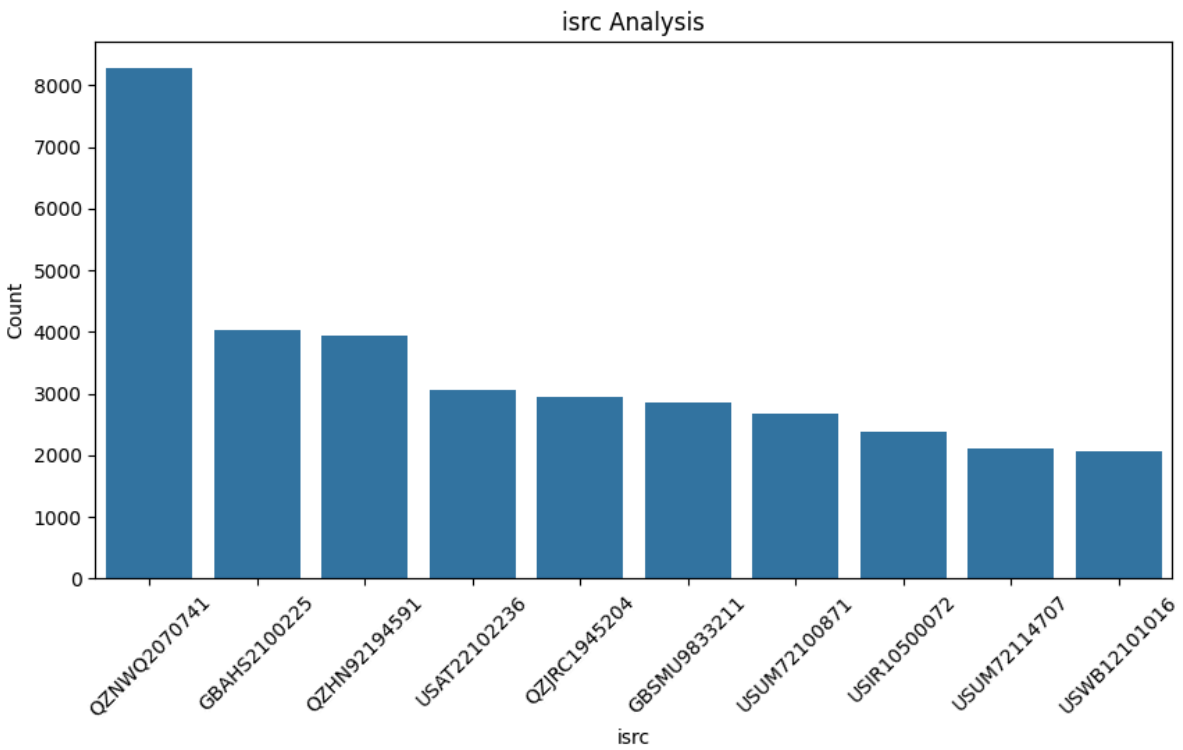
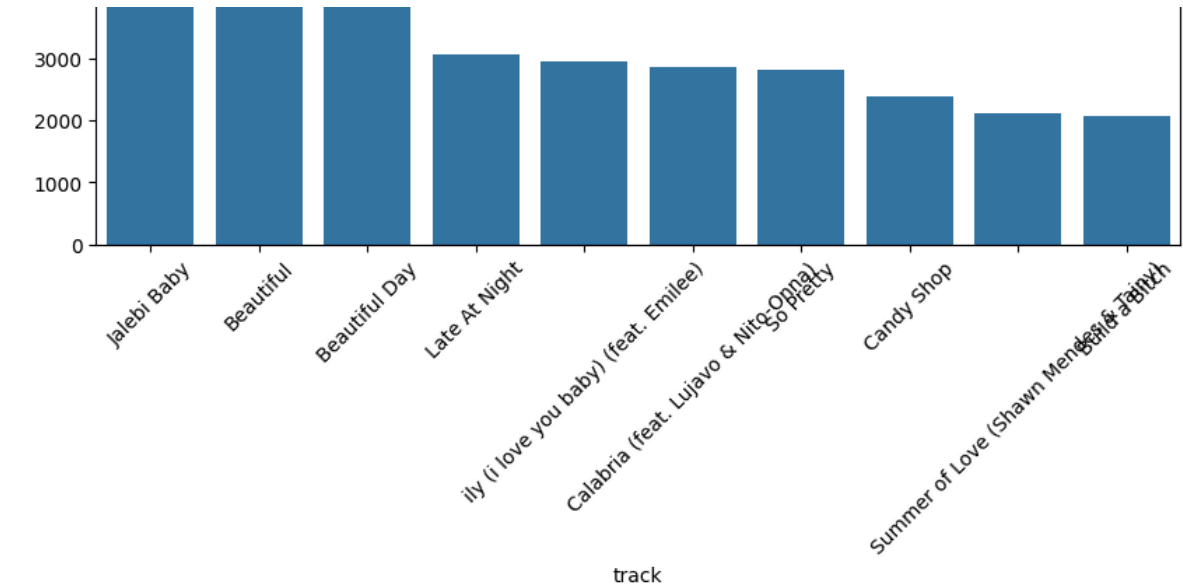
```
USUM72114707      2108
USWB12101016      2072
Name: count, dtype: int64
```

```
TOP VALUES IN linkid
linkid
2d896d31-97b6-4869-967b-1c5fb9cd4bb8      8288
e849515b-929d-44c8-a505-e7622f1827e9      3950
522da5cc-8177-4140-97a7-a84fdb4caf1c      3786
a82e72d8-523f-43ab-9fd0-992605d54b4e      3059
681d888a-59ce-4acb-b7c5-95dab0c063d9      2865
23199824-9cf5-4b98-942a-34965c3b0cc2      2675
c2c876ab-b093-4750-9449-6b4913da6af3      2442
97105aff-3e40-4c2e-bfeb-563b55e8f5dd      2390
d5317591-872a-487d-87cb-10a516643d5f      2074
a107c329-5d35-4c7d-8050-50bc69ce7dbc      2072
Name: count, dtype: int64
```





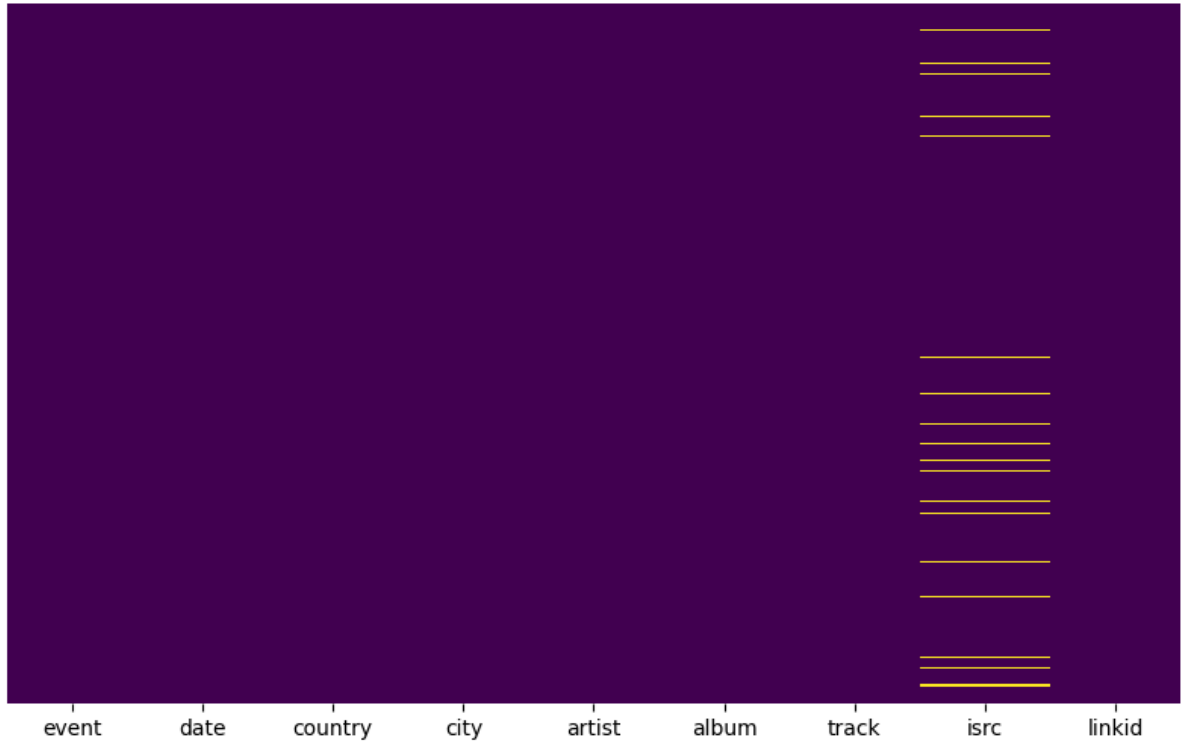




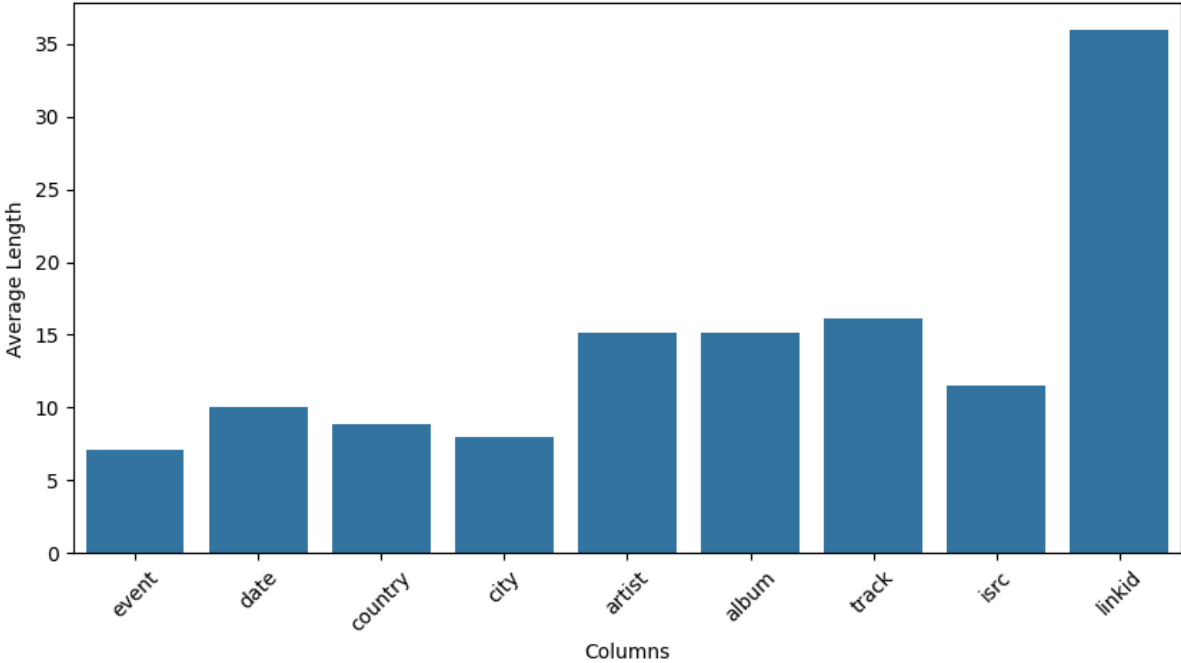
2d896d31-97b6-4869-967b-1c5fb-
e849515b-929d-44c8-a505-e7622-
522da5cc-8177-4140-97a7-a84fo-
a82e72d8-523f-43ab-9fd0-99260-
681d888a-59ce-4acb-b7c5-95dab-
23199824-9cf5-4b98-942a-34965-
c2c876ab-b093-4750-9449-6b491-
97105aff-3e40-4c2e-bfeb-563b5-
d5317591-872a-487d-87cb-10a51-
a107c329-5d35-4c7d-8050-50bc6-

linkid

Missing Values Heatmap



Average Text Length by Column



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Website Traffic Analysis

Objective

The objective of this project is to analyze website traffic data to identify user behavior, traffic patterns, and important trends using Python-based data analysis and visualization techniques.

Tools & Technologies Used Python Pandas Matplotlib Seaborn Jupyter Notebook
Data Cleaning Process Imported the dataset using Pandas Checked dataset information and missing values Removed duplicate records Cleaned column names for better analysis Exploratory Data Analysis

The dataset was analyzed using:

Value count analysis Distribution analysis Category-wise traffic analysis Missing value analysis Visualizations Used Bar Charts Count Plots Heatmaps
Key Insights Certain categories appeared more frequently in the dataset. Data distribution patterns were identified successfully. Missing values and duplicate records were analyzed. Visualizations helped understand the structure and trends of the dataset. Business Recommendations Focus on high-performing categories for better engagement. Improve low-performing areas using targeted strategies. Regular data cleaning should be performed to maintain data quality. Data visualization should be used frequently for business decision-making. Conclusion

The project successfully performed data cleaning, exploratory data analysis, and visualization. The analysis provided meaningful insights and satisfied the internship requirements for practical business-oriented data analysis.